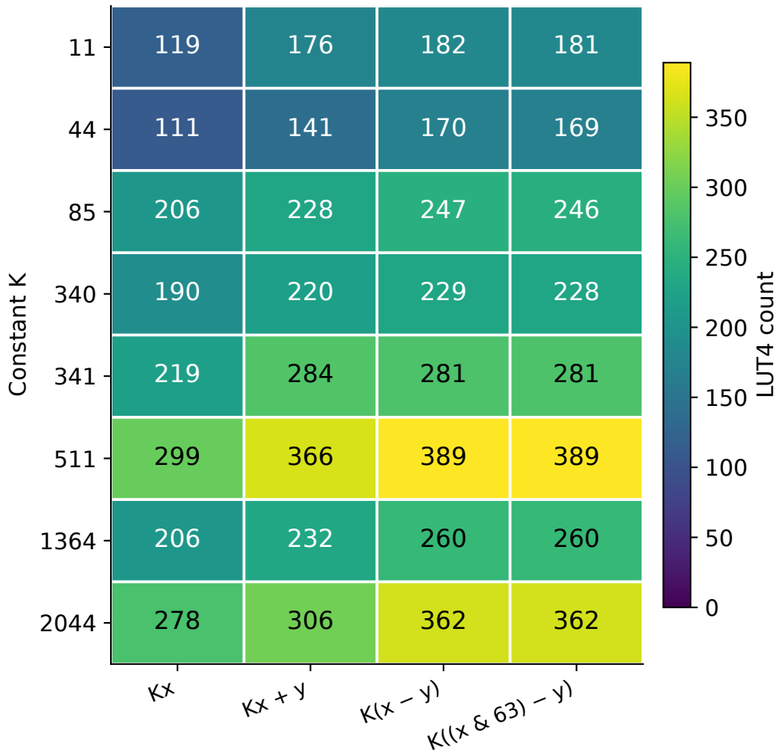
V10 · fixed CSD implementation versus original RTL

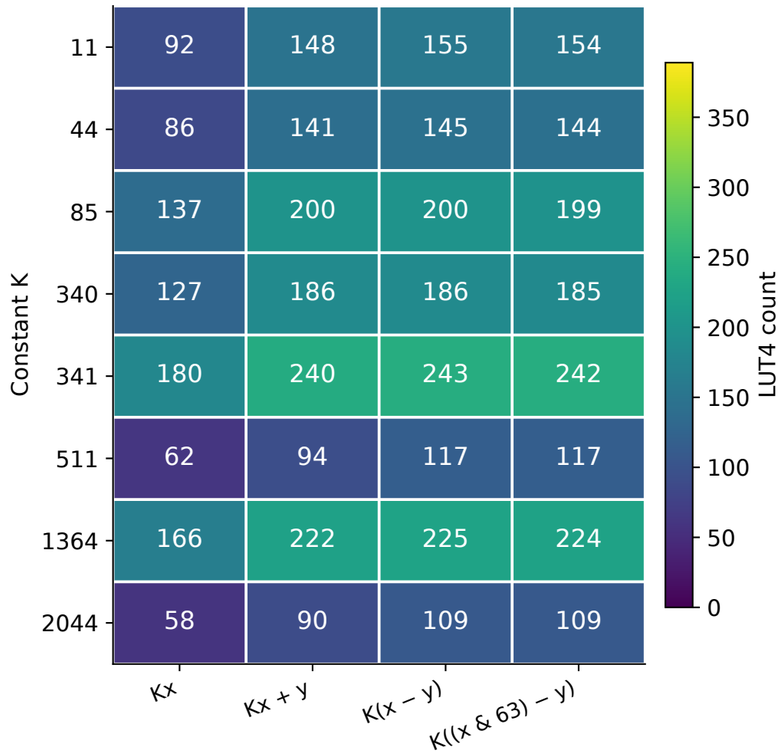
All 32 fixed constant/context pairs · 64 fresh original/CSD synthesis arms · no predictor fit or circuit selection

Wrapping unsigned32 arithmetic · isolated iCE40 LUT4 resource proxy · consumed development study

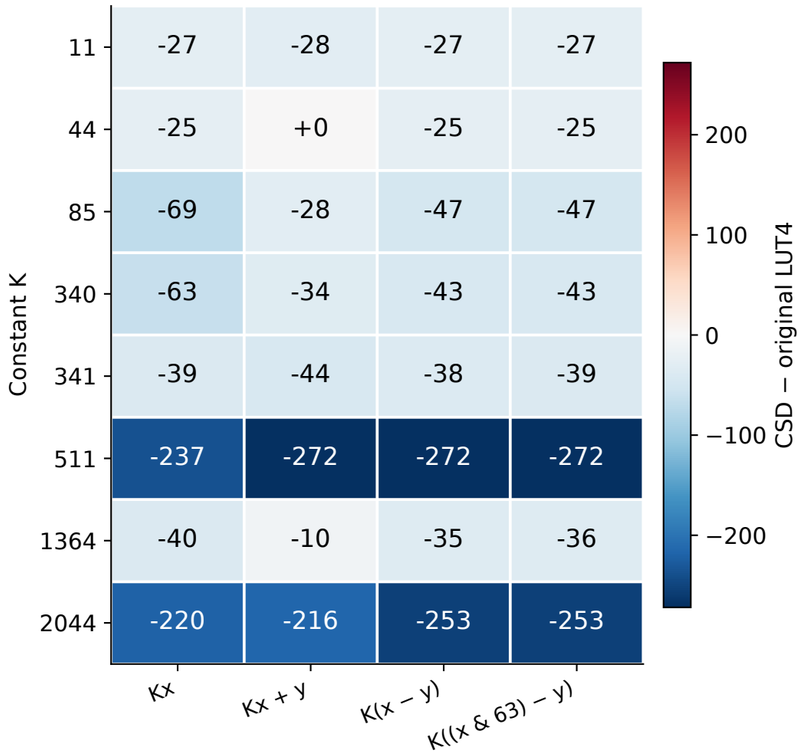
A Original LUT4



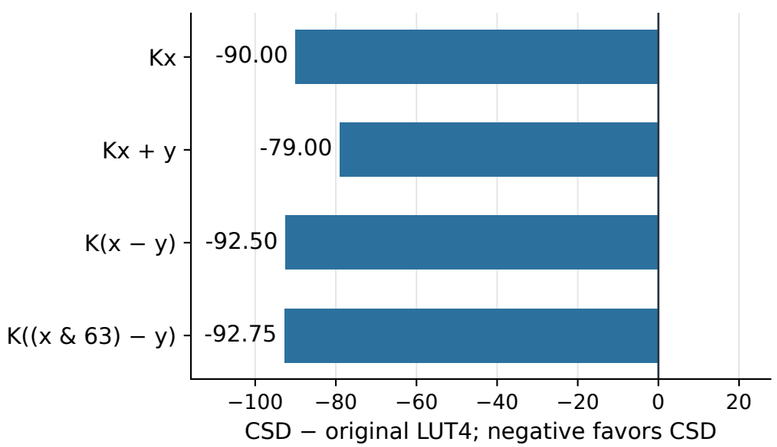
B CSD LUT4



C Paired Δ LUT4 · CSD − original



D Descriptive context means · 8 pairs each



E Primary LUT4 · carry descriptive

All 32 pairs	LUT4	Carry
Original mean	244.28	45.41
CSD mean	155.72	42.44
Mean paired Δ	-88.56	-2.97
Median paired Δ	-39.50	0.00
Minimum / maximum Δ	-272 / +0	-29 / +3
Wins / ties / losses	31 / 1 / 0	13 / 7 / 12

F Verification and scope

32 universal source QF_BV proofs
32 universal actual-RTL pair proofs
64 fresh synthesis arms, all retained

Per arm: 100 boundary + 2,048 seeded samples at RTL and mapped netlist. Mapped verification is sampled.

No population confidence interval; no device timing or CPU speedup claim.

x = in0; y = in1. Negative Δ reduces LUT4. Original and CSD heatmaps share a zero-based scale; paired Δ uses a symmetric scale.
All fixed pairs and losses shown. Full formulas, both arm counts and evidence hashes are in figure-data.json / figure-manifest.json.